

CMS Hospital Quality Analysis Change Log

I. Hospital_General_Information

A. Check for:

1. Null data that would affect analysis
 - a) Column K (Meets_criteria_for_birthing_friendly_designation), null indicates No
2. Misspelled words
 - a) No misspelled Words observed
3. Mistyped numbers
 - a) No mistyped Words observed
4. Extra spaces/characters
 - a) No extra characters observed
5. Duplicates
 - a) No duplicates found in data
6. Inconsistent/mismatched data formats and types
 - a) Column A (Facility ID) for Veteran's Health and Department of Defense (DoD) ID's include an F, so column is a string text
7. Misleading column names
 - a) No misleading column names
8. Truncated/missing data
 - a) DoD hospitals do not receive ratings so they will not be included in the analysis
9. Business Logic
 - a) Dataset includes hospitals from US territories including:
 - (1) 2 from the U.S. Virgin Islands (VI), which have ratings
 - (2) 1 from American Samoa (AS), which have no ratings
 - (3) 2 From Guam (GU) 1 of which has ratings in select measures
 - (4) 1 from Northern Mariana Islands (MP) with ratings in select measures (no overall rating)
 - (5) 61 from Puerto Rico (PR), 50 of which have ratings for select measures
 - (6) Total: 67 US territory hospitals, 54 of which have ratings for select measures

B. Updated field names

1. Description: Replaced all spaces and slashes with “_” in field names in Google Sheets
2. Date: 6/6/2026
3. Reason: Table cannot be imported into BigQuery with slashed in column name. Additionally, removing spaces will allow me to reference field names in queries without requiring quotations
4. Result: Field names all able to be imported and queried

C. Deleted footnote columns

1. Description: Removed the following footnote columns in Google Sheets
 - a) Hospital_overall_rating_footnote
 - b) MORT_Group_Footnote
 - c) Safety_Group_Footnote
 - d) READM_Group_Footnote
 - e) Pt_Exp_Group_Footnote
 - f) TE_Group_Footnote

2. Date: 6/6/2026
 3. Reason: Columns provided no meaningful data for future analysis
 4. Result: Shifted Last column from AL to AF
- D. Deleted Telephone Number Column*
1. Description: Deleted Column H (Telephone_numbers) in Google Sheets
 2. Date: 6/6/2026
 3. Reason: Not needed for analysis. Removed to make .csv file size smaller then 100MB for import to BigQuery
 4. Result: Shifted last column from AF to AE
- E. TRIM(facility_name)*
1. Description: Applied TRIM to facility_name column in SQL Query
 2. Date: 6/9/2026
 3. Reason: Removes space character at beginning or end of string that could potentially affect analysis
 4. Result: Extra spaces removed from all facility_name values
- F. CAST(hospital_overall_rating AS FLOAT64)*
1. Description: Applied CAST to hospital_overall_rating column in SQL to change the data type from text string to float
 2. Date: 6/9/2026
 3. Reason: Changing to numeric will allow proper calculations, aggregations, filtering, and ordering in later analysis
 4. Result: Data type changed to float in hospital_overall_rating column
- G. SAFE_CAST on all measures columns*
1. Description: Applied SAFE_CAST in SQL to the following measures column to change the data type from text string to integer
 - a) Count_of_MORT_Measures_Worse
 - b) Count_of_Safety_Measures_Worse
 - c) Count_of_READM_Measures_Worse
 - d) Count_of_MORT_Measures_Better
 - e) Count_of_Safety_Measures_Better
 - f) Count_of_READM_Measures_Better
 2. Date: 6/9/2026
 3. Reason: Changing to integer will allow proper calculations, aggregations, filtering, and ordering in later analysis. SAFE_CAST returns 'Not Available' values as NULL, excluding them from calculations without filtering out the entire row
 4. Result: Number values changed to integer data type and 'Not Available' values changed to NULL in measures columns
- H. hospital_overall_rating <> 'Not Available'*
1. Description: filtered hospital_overall_rating column in SQL to exclude rows where there is no data for hospital ratings
 2. Date: 6/9/2026
 3. Reason: This removes hospitals that do not have relevant data for analysis
 4. Result: All hospital rows with no available hospital rating removed from table. Row count changed from 5,426 to 2,866

II. HCAHPS_Hospital

- A. Check for:
1. Null data that would affect analysis
 - a) 'Not Available' and 'Not Applicable values in Patient_Survey_Star_Rating. Only 'Not Available' is present when filtered for 'summary star rating' in HCAHPS_Question column. After

reviewing data, rows are marked 'Not Available' if there are less than 100 surveys collected

2. Misspelled words
 - a) No misspelled words observed
3. Mistyped numbers
 - a) No mistyped numbers observed
4. Extra spaces/characters
 - a) No extra characters observed
5. Duplicates
 - a) No duplicate values found
6. Inconsistent/mismatched data formats and types
 - a) Facility ID column same as Hospital_General_Information table
7. Misleading column names
 - a) None observed
8. Truncated/missing data
 - a) None Observed
9. Business Logic
 - a) Dataset includes hospitals from US territories including:
 - (1) 2 from the U.S. Virgin Islands (VI), 1 with ratings
 - (2) 57 from Puerto Rico (PR), 10 with ratings
 - (3) 1 from American Samoa (AS), no data
 - (4) 2 from Guam (GU), no data
 - (5) 1 from Northern Mariana Islands (MP), no data
 - (6) Total: 63 US territory hospitals, 11 with survey ratings

B. Updated field names

1. Description: Replaced all spaces and slashes with “_” in field names in Google Sheets
2. Date: 6/6/2026
3. Reason: Table cannot be imported into BigQuery with slashed in column name. Additionally, removing spaces will allow me to reference field names in queries without requiring quotations
4. Result: Field names all able to be imported and queried

C. Deleted footnote columns

1. Description: Removed the following footnote columns in Google Sheets:
 - a) Patient_Survey_Star_Rating_Footer
 - b) HCAHPS_Answer_Percent_Footer
 - c) Number_of_Completed_Surveys_Footer
 - d) Survey_Response_Rate_Percent_Footer
2. Date: 6/6/2026
3. Reason: Columns provided no meaningful data for future analysis
4. Result: Shifted Last column from V to R

D. Deleted Telephone Number Column

1. Description: Deleted Column H (Telephone_numbers) in Google Sheets
2. Date: 6/6/2026
3. Reason: Not needed for analysis. Removed to make .csv file size smaller than 100MB for import to BigQuery
4. Result: Shifted last column from R to Q

E. Deleted Start Date and End Date

1. Description: Deleted start_date and End_date columns in Google Sheets
2. Date: 6/6/2026

3. Reason: All values in these columns include the same data indicating survey results are included from 4/1/24 to 3/31/25. Notated this in Project Notes document
4. Result: Shifted last column from Q to O

F. TRIM(facility_name)

1. Description: Applied TRIM to facility_name column in SQL Query
2. Date: 6/9/2026
3. Reason: Removes space character at beginning or end of string that could potentially affect analysis
4. Result: Extra spaces removed from all facility_name values

G. CAST(patient_survey_star_rating AS FLOAT64)

1. Description: Applied CAST to patient_survey_star_rating column in SQL to change the data type from text string to float
2. Date: 6/9/2026
3. Reason: Changing to numeric will allow proper calculations, aggregations, filtering, and ordering in later analysis
4. Result: Data type changed to float in patient_survey_star_rating

H. HCAHPS_Question = 'Summary star rating', Patient_Survey_Star_Rating NOT IN ('Not Available','Not Applicable')

1. Description: Filtered in SQL on HCAHPS_question for 'summary star rating' value and on patient_survey_star_rating to exclude 'Not Available','Not Applicable'
2. Date: 6/9/2026
3. Reason: Filtering for a single HCAHPS question returns one row per hospital, rather than multiple rows for each question. This will allow for proper joins and result in distinct rows. Filtering to exclude 'Not Available' and 'Not Applicable' values removes facilities without proper survey data for analysis
4. Result: Distinct facility rows and only integer values in patient_survey_star_rating columns. Row Count changed from 325,652 to 3,183